$\text{co-cw} = T_{hi} - T_{c_1} = T_1$ $\quad \Delta T_2 = T_{ho} - T_{co}$

$\text{count-cu} = \Delta T_1 = T_{hi} - T_{c_o}$ $\quad \Delta T_2 = T_{ho} - T_{c_i}$

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

<u>Exam #2B</u>

Only one sheet of paper 81/2 x 11 is allowed

No phones or i-pads or computers are allowed

Duration: 70 minutes --- November 4, 2025

All Problems must be done

Problem 1: (40 points)

A double-pipe heat exchanger is to be designed to cool 5.5 gal/min of a hot oil from 250°F to 120°F using 11 gal/min of cooling water available at 70°F. The exchanger is to consist of 12-ft-long sections of ¾-in.-diameter copper tubing placed inside 1½-in.-diameter, well-insulated tubing, with water flowing in the annular space between the two tubes. How many sections of copper tubing are required if the flow of the two streams is (a) counter-current and (b) co-current? The overall heat transfer coefficient is U_o = 110 BTU/h ft² °F, the heat capacity of the oil is 0.55 BTU/lb °F, and the density of the oil is 52 lb/ft³. Assume that there is no heat transfer to the surrounding.

105.75 °F T_c

$C_{p\,water} = 4.18\ J/kg\cdot K = 1\ BTU/lb\cdot F$

$\rho_{water} = 1000\ kg/m^3 = 62\ lb/ft^3$

heat exchanger

$Q = m_c C_p (T_{oc} - T_{ic})$

$m_h C_p (T_{ih} - T_{oh})$

$Q = UA \dfrac{\Delta T_2 - \Delta T_1}{\ln\left(\frac{\Delta T_2}{\Delta T_1}\right)}$

$A = 2\pi r L \cdot N_t = \pi D L \cdot N_t$

unit conversion $\quad m \quad \rho$

Q via oil: $Q = \left(5.5 \frac{\cancel{gal}}{min} \cdot .133 \frac{\cancel{ft^3}}{\cancel{gal}}\right)\left(52 \frac{\cancel{lb}}{\cancel{ft^3}}\right)\left(.55 \frac{Btu}{\cancel{lb°F}}\right)(250-120)\cancel{°F} = 2719.72 \frac{Btu}{min}$

T w/ Q_{H_2O}: $2719.72 \frac{Btu}{min} = \left(11 \frac{\cancel{gal}}{min} \cdot .133 \frac{\cancel{ft^3}}{\cancel{gal}}\right)\left(62 \frac{\cancel{lb}}{\cancel{ft^3}}\right)\left(1 \frac{Btu}{lb°F}\right)(T_{o,c} - 70°F) \Rightarrow T_{o,c} = 99.98°F$

$Q_{min} \to Q_{hr}$: $2719.72 \frac{Btu}{min} \cdot 60 \frac{min}{hr} = 163183 \frac{Btu}{hr}$

a) counter

$163183 \frac{\cancel{Btu}}{\cancel{hr}} = 110 \frac{\cancel{Btu}}{\cancel{hr}\, ft^2 \cancel{°F}} A \left(\dfrac{(120-99.98)-(250-70)\ \cancel{°F}}{\ln\left(\frac{120-99.98}{250-70}\right)}\right) \to A = 20.37\ ft^2$

$20.37\,ft^2 = \pi D L \cdot N_T$; $\frac{3}{4}\,in \cdot \frac{ft}{12\,in} = .0625\,ft$ (Assuming only inner tube, not outer tubing)

$20.37\,ft^2 = \pi(.0625\,ft)(12\,ft) \cdot N_T \rightarrow 2.356\,ft^2\,N_T \rightarrow \boxed{N_T = 8.64 \rightarrow 9 \text{ sections}}$

b) cocurrent

W/ co-current $\Delta T_2 : 250 - 99.98$

$\Delta T_1 : 120 - 70$

$$163183\,\frac{Btu}{hr} = 110\,\frac{Btu}{hft^2\,°F}\,A\left(\frac{(250-99.98)-(120-70)\,°F}{\ln\left(\frac{250-99.98}{120-70}\right)}\right)$$

$$163183\,\frac{Btu}{hr} = 10013.4\,\frac{Btu}{hft^2}\,A \rightarrow A = 16.2964\,ft^2$$

$$\pi D L \cdot N_T = 16.2964\,ft = \underbrace{\pi(.0625\,ft)(12\,ft)}_{2.356\,ft^2} N_T$$

$$\boxed{N_T = 6.92 \rightarrow 7 \text{ sections}}$$

Problem 2: **(40 points)**

Steel balls 10 mm in diameter are annealed by heating to 1150 K and then slowly cooling to 450 K in an air environment for which $T_\infty = 325$ K and $h = 25$ W/m$^2 \cdot$ K. Assuming the properties of the steel to be $k = 40$ W/m $\cdot$ K, $\rho = 7800$ kg/m^3, and $c = 600$ J/kg $\cdot$ K, estimate the time required for the cooling process.

$D = 10\,mm \to .01\,m$ $\quad T_1 = 1150\,K \quad T_2 = 450\,K \quad T_\infty = 325\,K \quad h = 25 \frac{W}{m^2} = \frac{J}{s m^2} \quad K = 40 \frac{W}{mK} = \frac{J}{smK}$

$\rho = 7800 \frac{kg}{m^3} \quad c = 600 \frac{J}{kgK} \quad A = 4\pi r^2 \quad V = \frac{4}{3}\pi r^3$

$-hA_s(T - T_\infty) = \rho V_c \frac{dT}{dt}$

$Bi: \frac{hL}{k} \to \frac{h}{k} L \to \frac{h}{k} \cdot \frac{V}{A} \to \frac{25 \frac{W}{Km^2}}{40 \frac{W}{mK}} \left(\frac{\frac{4}{3}\pi (.01)^3 m}{4\pi (.01)^2 m} \right) = .002 \ll 1$ So lumped model can be assumed

$-\frac{hA_s}{\rho V_c} dt = \frac{1}{(T - T_\infty)} dT \to \int_{T_i}^{T_F} \frac{1}{T - T_\infty} dT \to \ln\left(\frac{T_F - T_\infty}{T_i - T_\infty}\right)$

$-\frac{hA_s}{\rho V_c} t = \ln\left(\frac{T_F - T_\infty}{T_i - T_\infty}\right) \to t = -\frac{\rho V_c}{hA_s} \ln\left(\frac{T_F - T_\infty}{T_i - T_\infty}\right)$

$t = -\frac{7800 \frac{kg}{m^3} \left(\frac{4}{3}\pi \left(\frac{.01}{2} m\right)^3\right)}{25 \frac{W}{m^2 K} \left(4\pi \left(\frac{.01}{2} m\right)^2\right)} \ln\left(\frac{450 - 325}{1150 - 325}\right) = .981 \frac{kgk}{wm} = \frac{kgks}{J}$

$.981 \frac{kgks}{J} \cdot 600 \frac{J}{kgK} = 588.77$ seconds

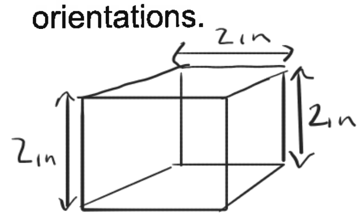
$\downarrow$

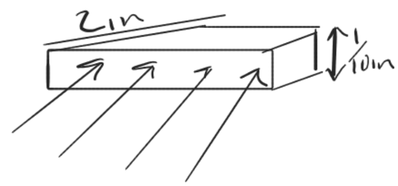
9.81 minutes

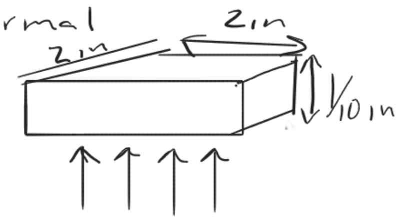
Problem 3: (30 points)

A cube 2.0-in. long on each side, is constructed of equal-sized, alternating parallel layers of copper and pyrex glass. There are ten layers of copper and fen layers of glass. A question has risen whether this cube would be a better conductor of heat (for a fixed overall ΔT) if it were oriented so that heat was conducted parallel to the alternating layers or normal to them. Find the ratio of the heat conduction rates for the two different orientations.

ratio of parallel/ normal







d^2T = constant

10 glass
10 copper
20 total layers

thickness 2in → $\frac{2}{20} = .1$ in, Know $q = KA\frac{dT}{dx}$ w/ $A = L \times W$

With normal → $q = KA\frac{dT}{dx} \to K_{glass}(2\times2)\frac{dT}{.1} \to 10q = 10(K_{glass}(2\times2)\frac{dT}{.1})$

w/ copper $q = K_{cu}(2\times2)\frac{dT}{.1} \to 10q = K_{cu}(2\times2)\frac{dT}{.1}10$

$q_{tot,n} = 10q + 10q \to k_{glass}\,4in^2\frac{dT}{.1} + K_{cu}\,4in^2\frac{dT}{.1} \to 10(4in^2)\frac{dT}{.1}(K_{glass}+K_{cu}) = q_{Tot,n}$

With parallel → $q = KA\frac{dT}{dx}$

glass: $10q = 10K_{glass}(2\cdot.1in)\frac{dT}{.1}$, w/ copper: $10q = 10K_{cu}(2\cdot.1in)\frac{dT}{.1}$

$q_{tot,p} = 10q + 10q \to K_{glass}(.2in^2)\frac{dT}{.1} + K_{cu}(.2in^2)\frac{dT}{.1}$ 2

$10(.2in^2)\frac{dT}{.1}(K_{glass}+K_{cu}) = q_{tot,p}$

$$\frac{\text{Parallel}}{\text{normal}} = \frac{q_{tot\,p}}{q_{tot\,n}} = \frac{\cancel{10}(.2in^2)\cancel{\frac{dT}{.1}}\cancel{(K_{glass}+K_{cu})}}{\cancel{10}(4in^2)\cancel{\frac{dT}{.1}}\cancel{(K_{glass}+K_{cu})}} = \frac{.2in^2}{4in^2} = \underline{.05}$$

if

$$\frac{\text{normal}}{\text{Parallel}} = \frac{10(4in^2)\frac{dT}{.1}(K_{glass}+K_{cu})}{10(.2in^2)\frac{dT}{.1}(K_{glass}+K_{cu})} = \frac{4in^2}{.2in^2} = \underline{20}$$